

## Supplementary Reading Material: On Dual Spaces

### Finite-Dimensional Vector Spaces and Dual Spaces

**Definition.** Let  $V$  be a vector space over a field  $\mathbb{F}$ . The dual space is defined as

$$V^* := \text{Hom}_{\mathbb{F}}(V, \mathbb{F}).$$

**Example.** For  $\dim V = n < \infty$ , with basis  $u_i$   $1 \leq i \leq n$ , the dual space  $V^*$  is also  $n$ -dimensional with basis (dual basis)

$$\{f_i\}_{1 \leq i \leq n}, \quad f_i : V \rightarrow \mathbb{F}, \quad u_j \mapsto \delta_{i,j}.$$

From the perspective of bilinear forms, the pairing

$$V \times V^* \rightarrow \mathbb{F}, \quad (u, f) \mapsto f(u)$$

has matrix form  $I_{n \times n}$  under the bases  $u_i$   $1 \leq i \leq n$  and  $f_i$   $1 \leq i \leq n$ .

**Note.** The isomorphism  $V \simeq V^*$  is artificial. An informal but illustrative analogy: assume  $V$  has unit kilometre (km), then  $V^*$  has unit  $\text{km}^{-1}$ . The linear isomorphism  $V \simeq V^*$  is never natural.

**Example.** The canonical linear map  $\eta_V : V \rightarrow V^{**}$  is a natural isomorphism.

- The term "natural" signifies that for any linear map  $f : U \rightarrow V$ , the compositions  $U \xrightarrow{f} V \xrightarrow{\eta_V} V^{**}$  and  $U \xrightarrow{\eta_U} U^{**} \xrightarrow{f^{**}} V^{**}$  coincide.

### Exotic Examples in Linear Algebra Without the Axiom of Choice

**Example.** ( $\dim V = \infty$ , yet  $V^* = 0$ ). If the axiom of choice is not assumed, then there do exist infinite-dimensional vector spaces  $V$  with dual space  $V^* = 0$ . The [example](#) by H. Läuchli presents an infinite-dimensional vector space  $V$  whose proper subspaces are all finite-dimensional.

**Example.** (When  $V^*$  is strictly larger than  $V$ , yet  $V \simeq V^{**}$ ). In ZF set theory with dependent choice, it takes effort to verify  $\mathbb{R}[[x]]^* \simeq \mathbb{R}[x]$ . However,  $\mathbb{R}[[x]]$  does not possess a countable basis.

**Proposition.** The canonical linear map  $V \rightarrow V^{**}$  is injective if and only if  $V$  is a subspace of some product space  $\prod_X \mathbb{F}$ . For instance,  $\mathbb{F}[x]$  and  $\mathbb{F}[[x]]$  both embed into their double dual spaces.

*Proof.* See [this article](#).

## Axiom of Extension

**Axiom.** (Axiom of Extension, also known as the Hahn–Banach Axiom (HB)).

- For any linear subspace  $U \subseteq V$ , every linear map  $f : U \rightarrow W$  extends to a map  $\tilde{f} : V \rightarrow W$  such that  $\tilde{f}|_U = f$ .

**Remark.** The Hahn–Banach Axiom is independent of the usual axioms (ZF + DC), which are assumed in the study of tensor products of infinite-dimensional vector spaces.

**Exercise.** The Hahn–Banach Axiom is equivalent to the assertion that  $\text{Hom}_{\mathbb{F}}(-, W)$  always maps injections to surjections.

**Exercise.** By HB, for any  $v_1, v_2 \in V$  with  $v_1 \neq v_2$ , there exists  $f \in V^*$  such that  $f(v_1) \neq f(v_2)$ . Hence, the canonical linear map  $V \rightarrow V^{**}$  is always injective.

**Exercise.** By HB, show that for  $U \hookrightarrow V$ , the following map is an isomorphism

$$\frac{V^{**}}{U^{**}} \rightarrow (V/U)^{**}, \quad [x]_{U^{**}} \mapsto \pi^{**}(x),$$

where  $\pi$  is the natural quotient map  $V \twoheadrightarrow V/U$ .

## Dual Spaces and Galois Correspondence

**Definition.** (Kernels and Annulators). For convenience, let  $U_i$  denote subspaces of  $V$ , and  $B_i$  subspaces of  $V^*$ .

- $\text{ann } U := \{f \in V^* \mid f(U) = 0\}$ , a subspace of  $V^*$ ;
- $\ker B := \{u \in V \mid f(u) = 0 \ (\forall f \in B)\} = \bigcap_{f \in B} \ker f$ , a subspace of  $V$ .

These notions are defined for subsets in general. By the closure property:

- $\text{ann}(X) = \text{ann}(\text{span}(X))$ , for  $X \subseteq V$ ;
- $\ker(S) = \ker(\text{span}(S))$ , for  $S \subseteq V^*$ .

**Remark.**  $\ker f$  and  $\ker f$  are identical.

**Exercise.** Show that for any finite-dimensional vector space  $V$ , the map  $\text{ann} : \text{Subsets of } V \rightarrow \text{Subsets of } V^*$  is a bijection with inverse map  $\ker$ .

**Proposition.** One has  $\text{ann}(\ker B) \supseteq B$ . Equality does not generally hold.

*Proof.* The formula  $\text{ann}(\ker B) \supseteq B$  involves logic more than algebra. For a counterexample, take  $V$  as the space of continuous functions over  $\mathbb{R}$ , and  $B = \text{span}(\{\delta_r\}_{r \in \mathbb{Q}})$ .

**Exercise.** (Galois Connection)

1.  $\text{ann}(\ker B) \supseteq B$ ,

2.  $\ker(\text{ann } U) \supseteq U$ ,
3.  $\text{ann}(\ker(\text{ann } B)) = \text{ann}(B)$ ,
4.  $\ker(\text{ann}(\ker B)) = \ker B$ .

The pair  $(\ker, \text{ann})$  defines a Galois correspondence.

**Example.** For a set map  $f : X \rightarrow Y$ , the image and preimage maps  $(f_*, f^*)$  form a Galois connection between  $\text{Subset}(X)$  and  $\text{Subset}(Y)$ .

**Definition.** Let  $Op(\mathbb{R}^2)$  denote the set of open subsets of  $\mathbb{R}^2$ , and  $Cl(\mathbb{R}^2)$  the set of closed subsets. Define:

1.  $k : Op(\mathbb{R}^2) \rightarrow Cl(\mathbb{R}^2), \quad U \mapsto \overline{U}$  (closure);
2.  $i : Cl(\mathbb{R}^2) \rightarrow Op(\mathbb{R}^2), \quad K \mapsto K^\circ$  (interior).

**Exercise.** Show that  $ikik = ik$  and  $kiki = ki$ ; hence,  $(k, i)$  form a Galois connection.

**Exercise.** Given  $X \subseteq \mathbb{R}^2$ , show that at most seven distinct sets (including  $X$ ) can be obtained by applying  $k$  and  $i$  repeatedly.

**Exercise.** Show that one can obtain at most 14 distinct sets (including  $X$  itself) by applying  $i$  and  $(-)^c$  repeatedly.

## The Yoga of Dual Spaces

**Theorem.** In the following table, FD denotes finite-dimensional cases, G denotes general cases, and HB refers to cases where the Hahn-Banach Axiom is assumed.

For a linear map  $\varphi : U \rightarrow V$ , define the dual map as

$$\varphi^* : V^* \rightarrow U^*, \quad [V \rightarrow \mathbb{F}, v \mapsto f(v)] \mapsto [U \rightarrow \mathbb{F}, u \mapsto f(\varphi(u))].$$

| No. | Formula                                                                      | FD | G           | HB          |
|-----|------------------------------------------------------------------------------|----|-------------|-------------|
| G1  | $\text{ann}(\ker B) \stackrel{?}{=} B$                                       | =  | $\supseteq$ | $\supseteq$ |
| G2  | $\ker(\text{ann } V) \stackrel{?}{=} V$                                      | =  | $\supset$   | =           |
| E1  | $\text{ann}(V_1 + V_2) \stackrel{?}{=} \text{ann}(V_1) \cap \text{ann}(V_2)$ | =  | =           | =           |
| E2  | $\text{ann}(V_1 \cap V_2) \stackrel{?}{=} \text{ann}(V_1) + \text{ann}(V_2)$ | =  | $\supseteq$ | =           |

| No. | Formula                                                                    | FD       | G                 | HB          |
|-----|----------------------------------------------------------------------------|----------|-------------------|-------------|
| E3  | $\ker(B_1 + B_2) \stackrel{?}{=} \ker B_1 \cap \ker B_2$                   | =        | =                 | =           |
| E4  | $\ker(B_1 \cap B_2) = \ker B_1 + \ker B_2$                                 | =        | $\supseteq$       | $\supseteq$ |
| Z1  | $\ker \varphi^* \stackrel{?}{=} \text{ann}(\text{im } \varphi)$            | =        | =                 | =           |
| Z2  | $\ker \varphi \stackrel{?}{=} \ker(\text{im } \varphi^*)$                  | =        | $\subseteq$       | =           |
| Z3  | $\text{im } \varphi^* \stackrel{?}{=} \text{ann}(\ker \varphi)$            | =        | $\subseteq$       | =           |
| Z3' | $\text{im } \pi^* \stackrel{?}{=} \text{ann}(\ker \pi)$ , with $\pi$ surj. | =        | =                 | =           |
| Z4  | $\text{im } \varphi = \ker(\ker \varphi^*)$                                | =        | $\subseteq$       | =           |
| Z5  | If $f$ is surj., then $f^*$ is inj.                                        | true     | true              | true        |
| Z6  | If $f$ is inj., then $f^*$ is surj.                                        | true     | NaN               | true        |
| L1  | $S^* \stackrel{?}{\rightarrow} V^* / \text{ann}(S)$                        | $\simeq$ | $\hookrightarrow$ | $\simeq$    |
| L2  | $(V/S)^* \stackrel{?}{\rightarrow} \text{ann}(S)$                          | $\simeq$ | $\simeq$          | $\simeq$    |

**Remark.** When the general case (G) coincides with the HB case, the proposition is provable without additional axioms. When the FD case differs from the G case, the FD part is proved using dimension arguments. Nonexamples disprove the HB part of statements G1 and E4.